📄 **Product Requirements Document (PRD)**

🚑 **PinResQ – Geo-Verified Emergency Response System**

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1. 🧠 Product Overview

PinResQ is a web-based emergency response platform that enables users to report accidents in real-time and allows nearby individuals to view and respond quickly using location-based data and map visualization.

The system focuses on reducing emergency response time by leveraging **geolocation, real-time reporting, and community participation**.

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2. 🎯 Objectives

Primary Goals:

- Enable **instant accident reporting**
- Provide **accurate GPS-based location tracking**
- Display incidents on a **live interactive map**
- Reduce delay in emergency response

Success Criteria:

- Report submission time < 5 seconds
- Map load time < 3 seconds
- Accurate location detection (>95%)
- System uptime > 99%

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3. 🚨 Problem Statement

- Emergency services are often delayed
- Lack of real-time reporting tools

- No visibility of nearby incidents
- Bystanders are not effectively utilized
- Fake or unclear reports reduce efficiency

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4. 💡 Solution Overview

PinResQ provides:

- One-click emergency reporting
- Automatic GPS location capture
- Backend data storage and retrieval
- Map-based visualization of incidents
- Community visibility for faster response

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5. 👤 Target Users

User Type	Description
General Users	Report accidents
Nearby Users	View and respond to incidents
Admin (future)	Monitor system

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6. 🔑 Core Features

6.1 User Authentication

- Register with name, email, password
- Secure login system
- Basic session handling

6.2 Emergency Reporting

- Button-based reporting
- Auto-fetch latitude & longitude
- Store report in database

6.3 Map Visualization

- Interactive map using Leaflet
- Display markers for all reports
- Click markers for details

6.4 Backend API System

- REST APIs for communication
- Handles report creation and retrieval

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7. 🔄 User Flow

1. User opens application
2. Logs in / registers
3. Navigates to dashboard
4. Clicks “Report Emergency”
5. Location is captured automatically
6. Data sent to backend
7. Stored in database
8. Map displays the report
9. Other users can view it

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8. 🧱 System Architecture

Frontend (React.js)

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REST API (Axios)

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Backend (Spring Boot)

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Database (MySQL)

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9. 🗄️ Data Requirements

Users Table

- id (Primary Key)
- name
- email
- password
- phone (optional)

Reports Table

- id (Primary Key)
- latitude
- longitude
- timestamp
- user_id (Foreign Key)

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10. 🌐 API Requirements

Authentication APIs

- POST /api/auth/register
- POST /api/auth/login

Report APIs

- POST /api/reports/create
- GET /api/reports/all

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11. 🎨 UI/UX Requirements

Design Principles:

- Clean and minimal UI
- Mobile responsive
- Easy navigation

Key Pages:

- Login Page
- Register Page
- Dashboard
- Report Page
- Map Page

Color Scheme:

- Red → Emergency
- Blue → Trust
- Green → Success

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12. ⚙️ Technical Requirements

Frontend:

- React.js (JavaScript only)
- Axios for API calls
- React Router for navigation

- Leaflet for maps

Backend:

- Java Spring Boot
- RESTful APIs
- MVC architecture

Database:

- MySQL

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13. 🔒 Security Requirements

- Password encryption (BCrypt recommended)
- Input validation
- CORS configuration
- Basic authentication handling

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14. ⚠️ Constraints

- Requires internet connection
- Depends on browser location permissions
- Initial version does not include real-time updates
- No native mobile app (web only)

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15. 🧪 Testing Requirements

- API testing (Postman or browser)
- Frontend UI testing
- Integration testing (frontend + backend)

- Location accuracy testing

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16. 📈 Future Enhancements

- JWT authentication
- Image/video upload for verification
- Real-time notifications (WebSockets)
- Nearby volunteer detection (radius logic)
- Admin dashboard
- Mobile app (React Native)

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17. 📊 KPIs (Key Performance Indicators)

- Number of reports submitted
- Average response time
- Active users
- Map interaction rate

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18. 🏁 Conclusion

PinResQ is a scalable, real-world impactful web application that combines:

- Location intelligence
- Full-stack architecture
- Community-driven response

It lays the foundation for a **next-generation emergency response system**.

✅ You can now use this PRD for:

- 📊 PPT / Presentation

- 📄 Project Report
- 👨‍💻 Development roadmap
- 🧠 AI prompts (code generation)